

CarotisAi – Carotid Artery Analysis Report

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Image: demo_carotid_ultrasound.jpg
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& DISCLAIMER

DISCLAIMER: This AI analysis is a research/educational tool only. It does NOT constitute a clinical diagnosis. All findings must be verified by a qualified radiologist.

OVERALL RISK ASSESSMENT

Risk Category:	HIGH
AI Confidence:	high
Analysis Quality:	excellent

STENOSIS ANALYSIS

Grade:	MODERATE – 55–65%
Method:	NASCET criteria (visual estimate)
Side:	left
Haemodynamically Significant	No

PLAQUE ANALYSIS

Plaque Present:	YES
Location:	bifurcation
Side:	left
Echogenicity:	mixed
Texture:	heterogeneous
Shape:	irregular
Ulcerated:	No
Calcified:	Yes

Mixed echogenicity plaque with areas of calcification and soft components present at the carotid bifurcation extending into the proximal ICA. Irregular surface contour.

INTIMA-MEDIA THICKNESS (IMT)

Measured:	Yes
Value:	1.1 mm
Increased:	Yes

IMT measures 1.1 mm, exceeding the age-adjusted normal threshold of 0.9 mm, consistent with early atherosclerotic change.

FLOW ASSESSMENT

Flow Present:	Yes
Turbulent:	Yes
Reduced PSV:	No

Flow signal is present. Mild turbulence noted at the level of the plaque. Peak systolic velocity is within normal limits at this stage.

CLINICAL SUMMARY

Moderate stenosis (55–65% by NASCET) of the left internal carotid artery at the level of a mixed, calcified, heterogeneous plaque at the bifurcation. Increased IMT (1.1 mm) suggests progressive atherosclerosis. Haemodynamically not yet significant; however, the heterogeneous plaque morphology and irregular surface warrant close clinical follow-up. Cardiovascular risk stratification: HIGH.

RECOMMENDATIONS

1. Correlate with clinical symptoms (TIA, amaurosis fugax, focal neurological deficit).
2. Optimise medical therapy: statin, antiplatelet (if indicated), blood pressure control.
3. Repeat duplex ultrasound in 6 months to assess plaque progression.
4. Consider CT/MR angiography for further evaluation if symptoms develop or stenosis progresses beyond 70%.
5. Cardiology / Neurovascular multidisciplinary discussion recommended.

LIMITATIONS

This is a DEMO result generated from sample data — no real image was analysed. In clinical practice, image quality, patient habitus, and operator skill all influence measurement accuracy.